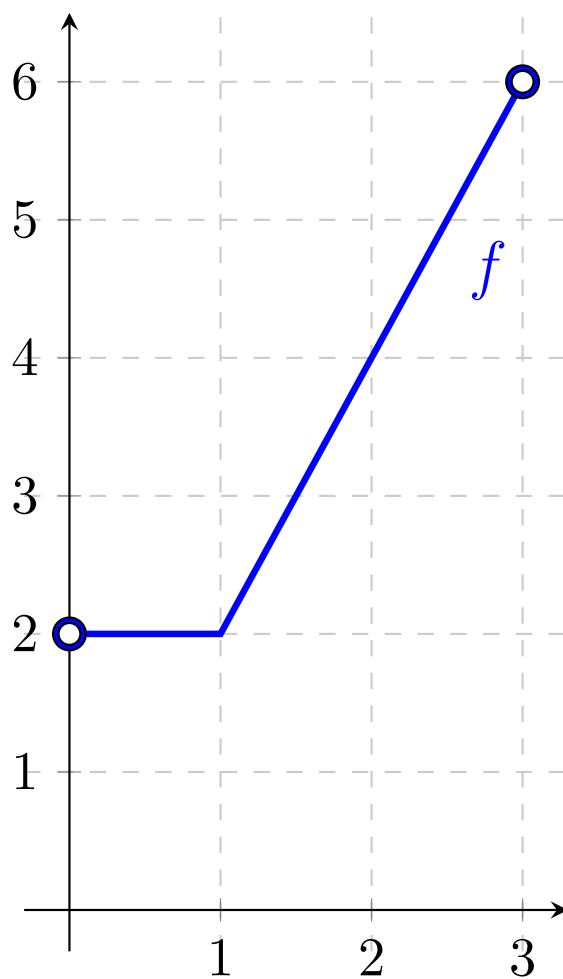


Problem 1

Problem 1 Find a formula for the function f whose graph is given in the figure below.

What is the domain of f ? What is the range of f ? Is the function f linear?



Explanation. $f(x) = \begin{cases} 2 & \text{if } 0 < x \leq 1 \\ 2 + 2(x - 1) & \text{if } 1 < x < 3 \end{cases}$

Domain of $f = (0, 3)$, range of $f = [2, 6)$.

f is not linear; because its graph is NOT a line; it is a piecewise defined function.